

# Engineering Curves: Construction of Cycloid

Rolling Circle Method with Tangent & Normal

---

Gajavada Sanjeevkumar

# Problem Statement

## Question

Draw a **cycloid** generated by a circle of diameter  $D = 50$  mm rolling along a straight base line for **one complete revolution** without slipping.

Construct a **tangent** and a **normal** to the cycloid at any point  $P$  located on the curve.

# Step 1: Draw Generating Circle & Directing Line (Base)

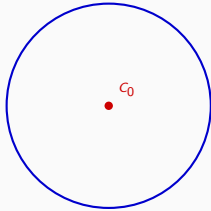
$C_0$  → Initial Center Point

$AA'$  → Directing Line

**Axis** → Center Line



# Step 1: Draw Generating Circle & Directing Line

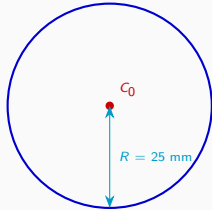


$C_0 \rightarrow$  Initial Center Point

$AA' \rightarrow$  Directing Line

**Axis**  $\rightarrow$  Center Line

# Step 1: Draw Generating Circle & Directing Line

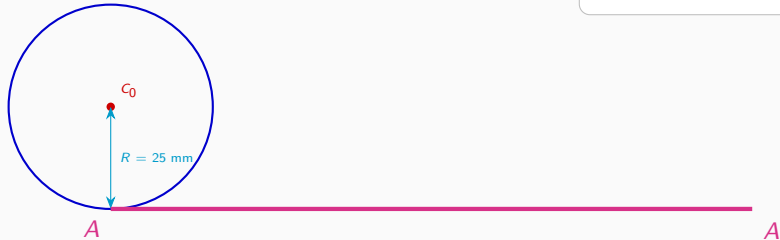


$C_0$  → Initial Center Point

$AA'$  → Directing Line

**Axis** → Center Line

# Step 1: Draw Generating Circle & Directing Line



$C_0 \rightarrow$  Initial Center Point  
 $AA' \rightarrow$  Directing Line  
**Axis**  $\rightarrow$  Center Line

# Step 1: Draw Generating Circle & Directing Line



$C_0 \rightarrow$  Initial Center Point  
 $AA' \rightarrow$  Directing Line  
**Axis**  $\rightarrow$  Center Line

# Step 1: Draw Generating Circle & Directing Line

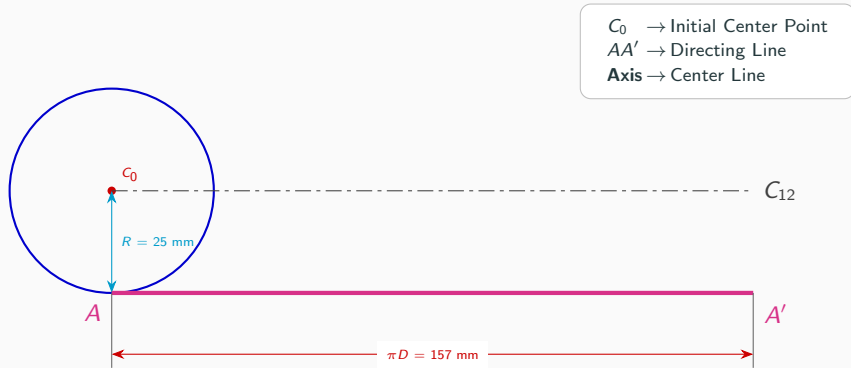


$C_0$  → Initial Center Point  
 $AA'$  → Directing Line  
**Axis** → Center Line

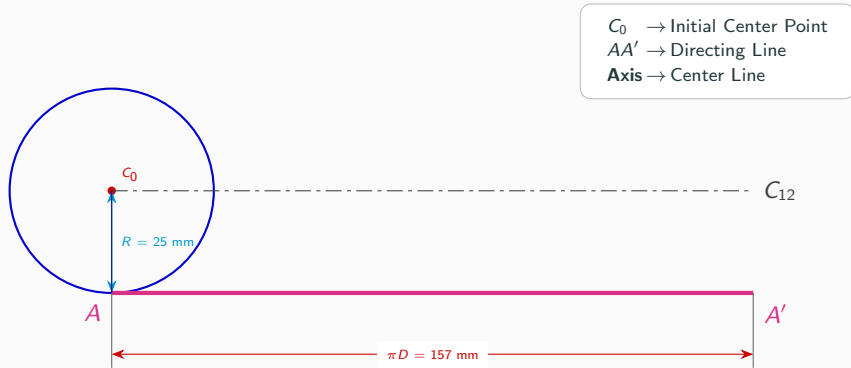
# Step 1: Draw Generating Circle & Directing Line



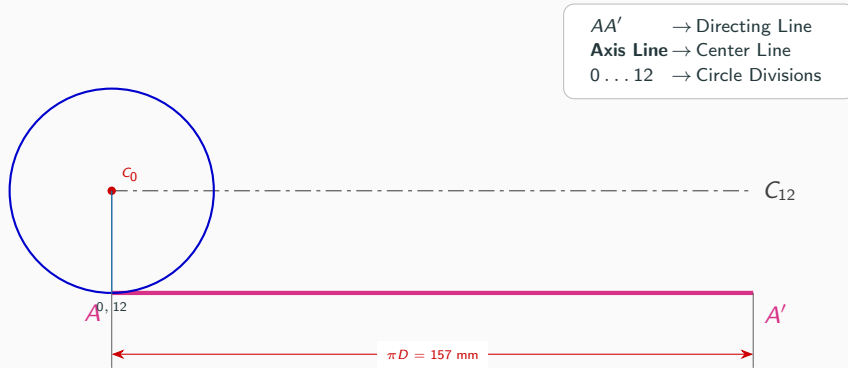
# Step 1: Draw Generating Circle & Directing Line



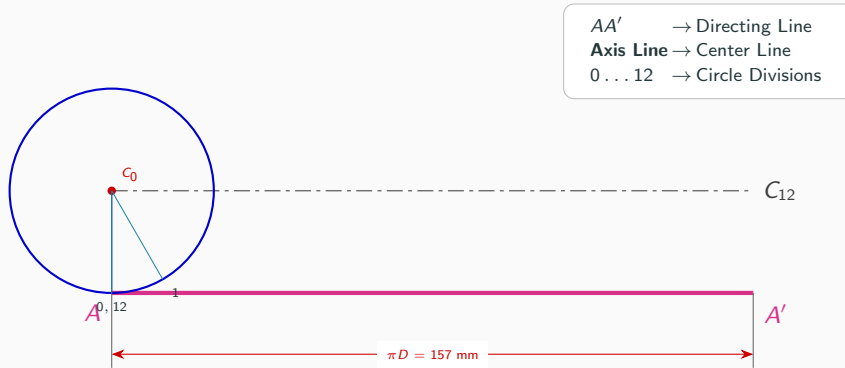
# Step 1: Draw Generating Circle & Directing Line



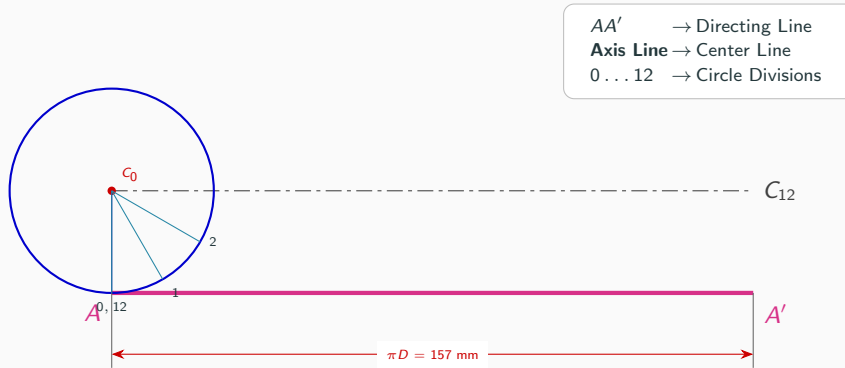
## Step 2: Divide Generating Circle into 12 Equal Parts



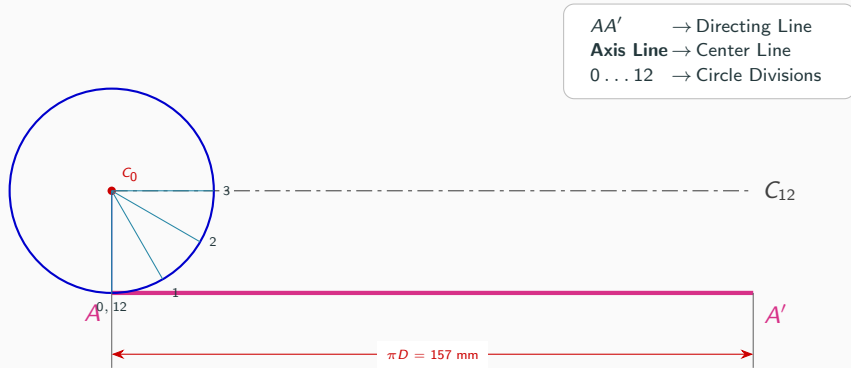
## Step 2: Divide Generating Circle into 12 Equal Parts



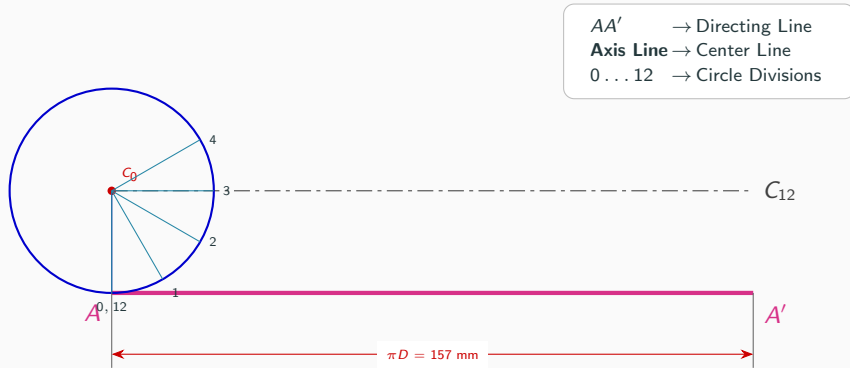
## Step 2: Divide Generating Circle into 12 Equal Parts



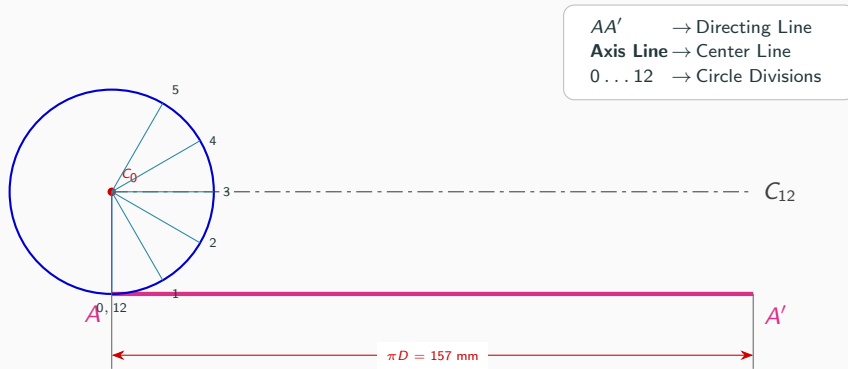
## Step 2: Divide Generating Circle into 12 Equal Parts



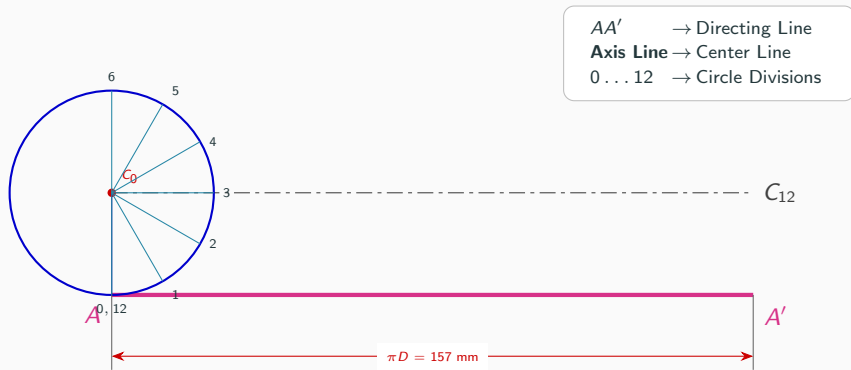
## Step 2: Divide Generating Circle into 12 Equal Parts



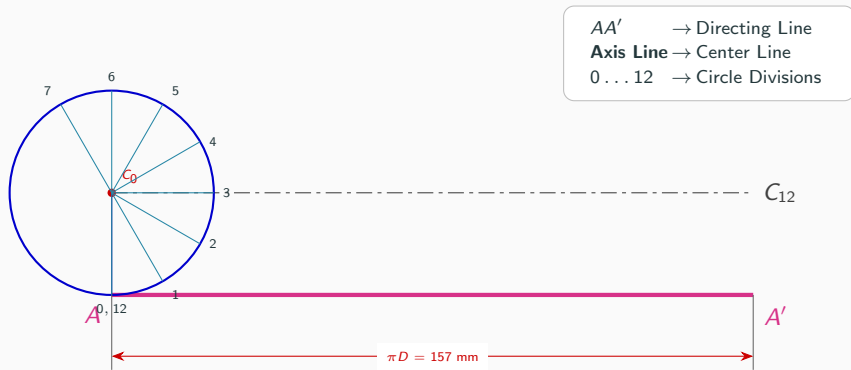
## Step 2: Divide Generating Circle into 12 Equal Parts



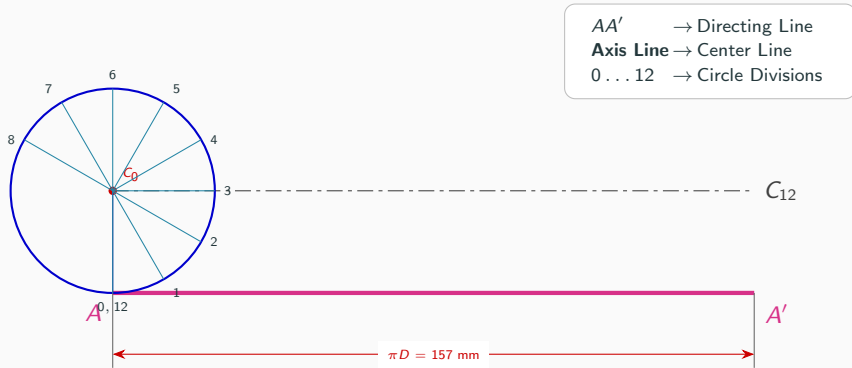
## Step 2: Divide Generating Circle into 12 Equal Parts



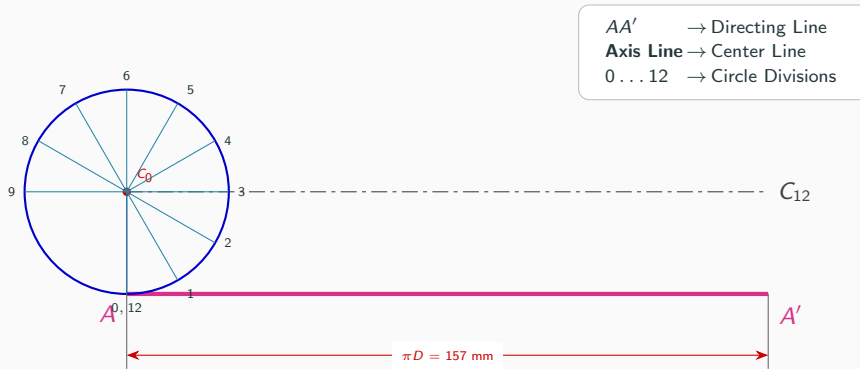
## Step 2: Divide Generating Circle into 12 Equal Parts



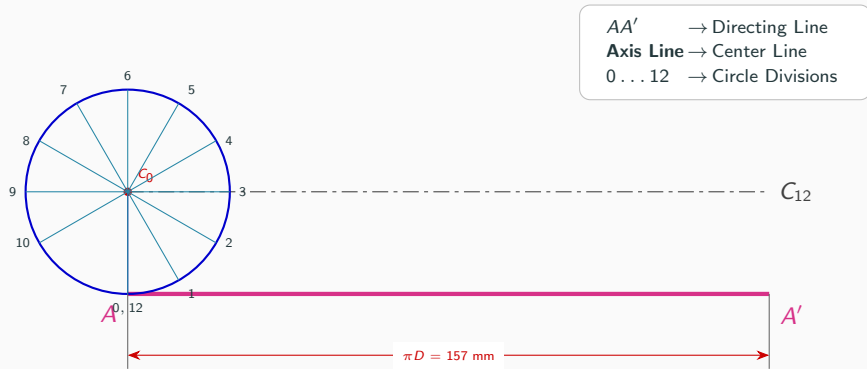
## Step 2: Divide Generating Circle into 12 Equal Parts



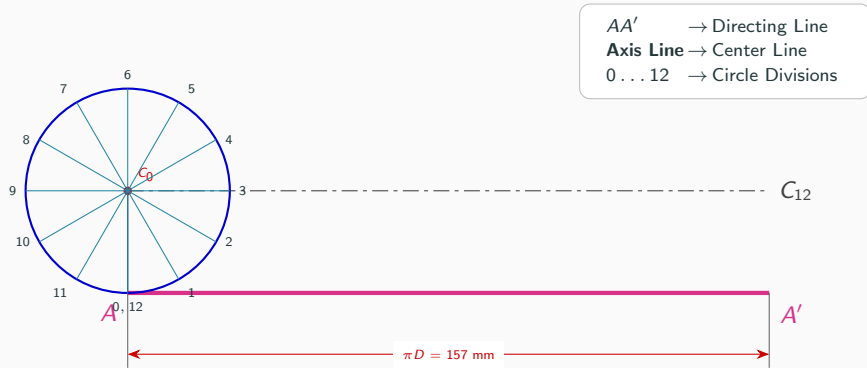
## Step 2: Divide Generating Circle into 12 Equal Parts



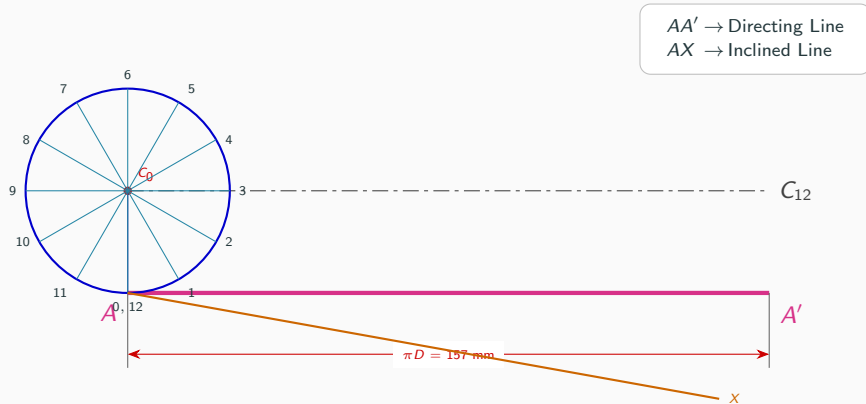
## Step 2: Divide Generating Circle into 12 Equal Parts



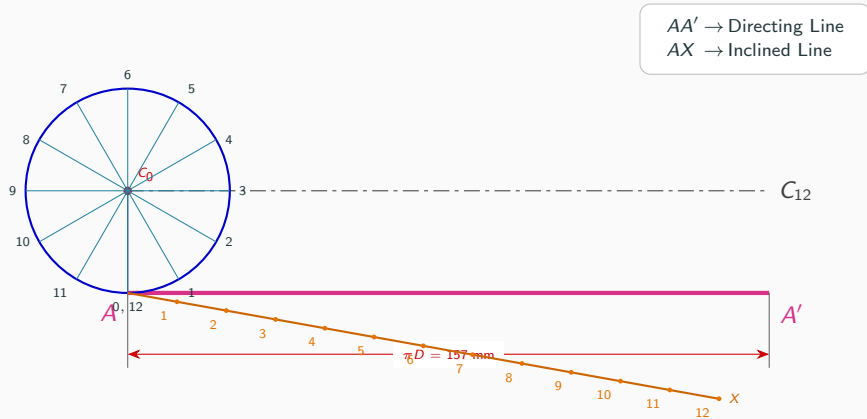
## Step 2: Divide Generating Circle into 12 Equal Parts



## Step 3: Divide Directing Line - Setup Inclined Line AX

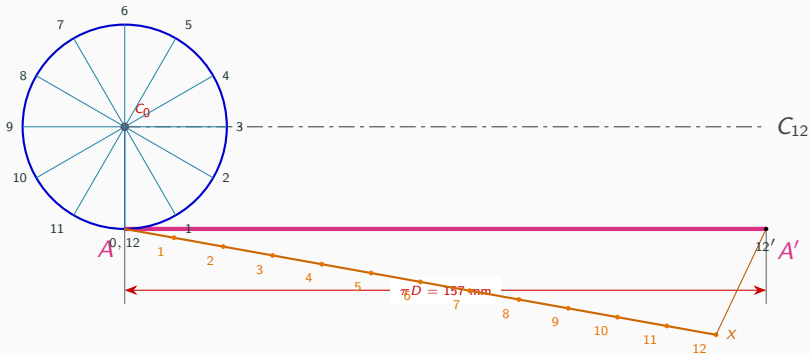


### Step 3: Divide Directing Line - Add Numbers on Line AX



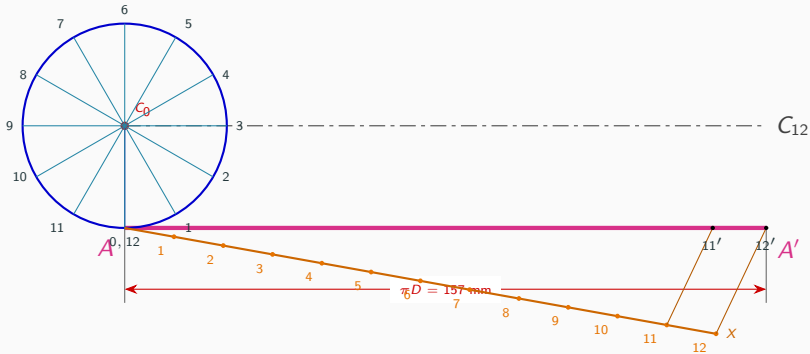
### Step 3: Divide Directing Line - Transfer Point 12'

12' → Transfer Point 12



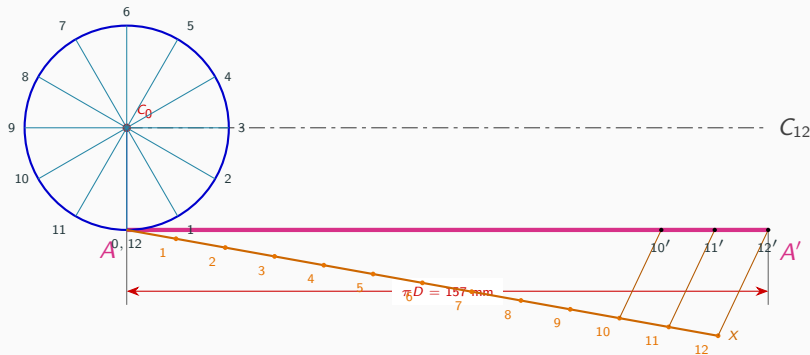
### Step 3: Divide Directing Line - Transfer Point 11'

11' → Transfer Point 11



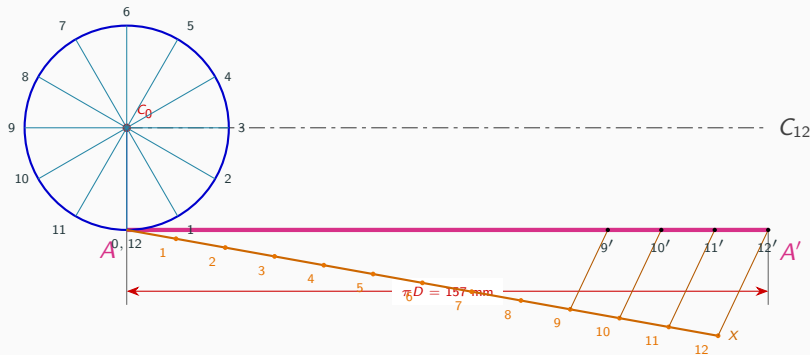
## Step 3: Divide Directing Line - Transfer Point 10'

10' → Transfer Point 10



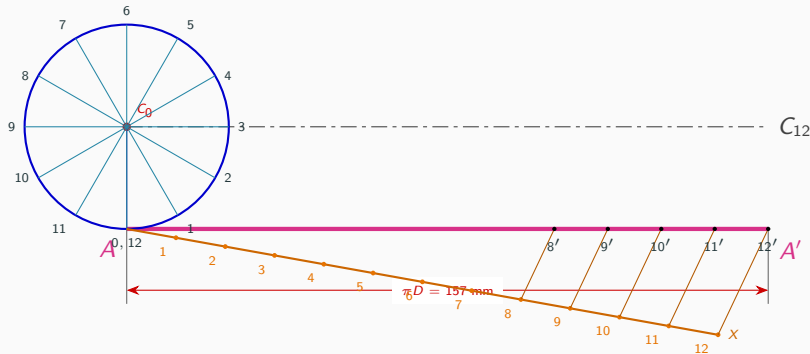
## Step 3: Divide Directing Line - Transfer Point 9'

9' → Transfer Point 9



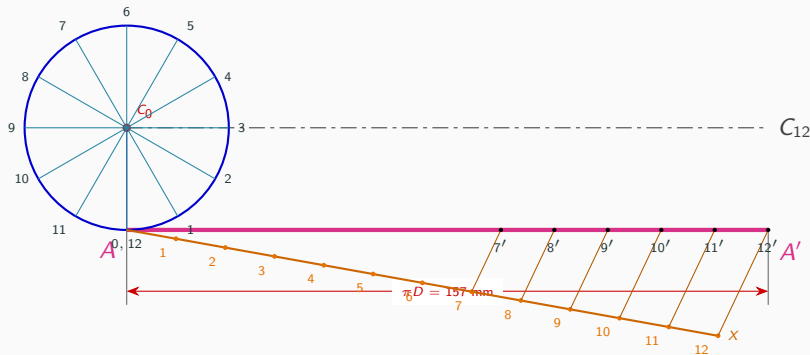
### Step 3: Divide Directing Line - Transfer Point 8'

8' → Transfer Point 8



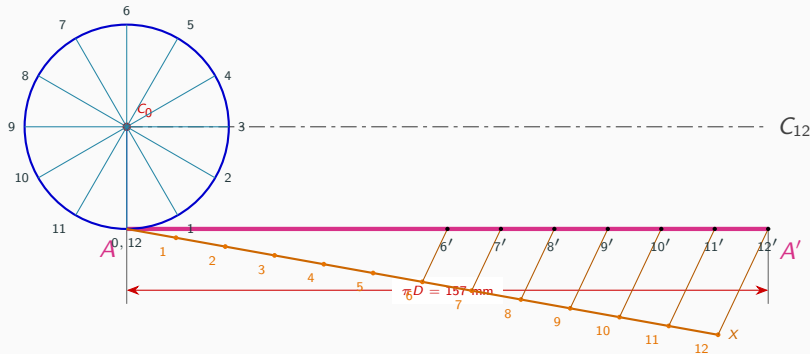
## Step 3: Divide Directing Line - Transfer Point 7'

7' → Transfer Point 7



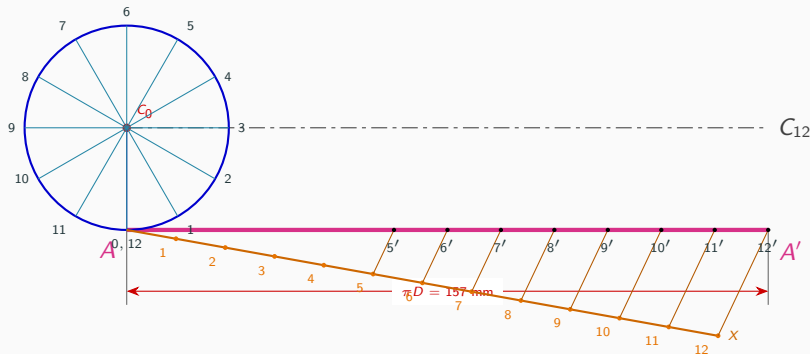
## Step 3: Divide Directing Line - Transfer Point 6'

6' → Transfer Point 6



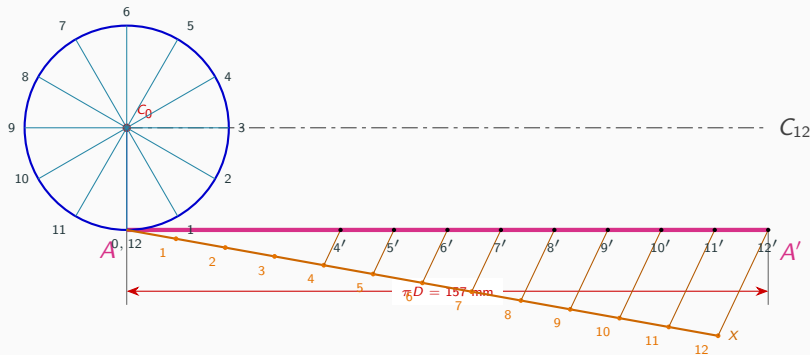
## Step 3: Divide Directing Line - Transfer Point 5'

5' → Transfer Point 5



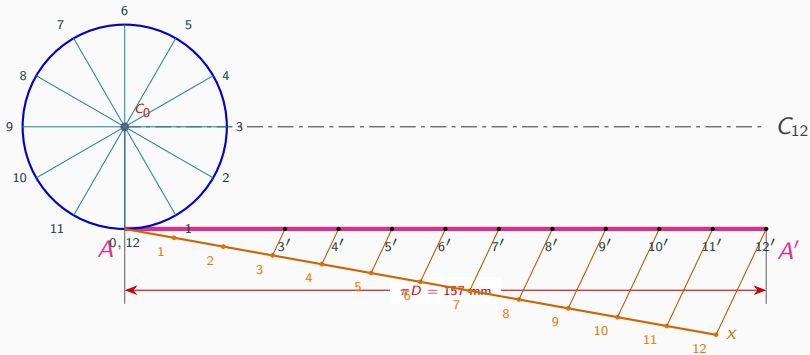
## Step 3: Divide Directing Line - Transfer Point 4'

4' → Transfer Point 4



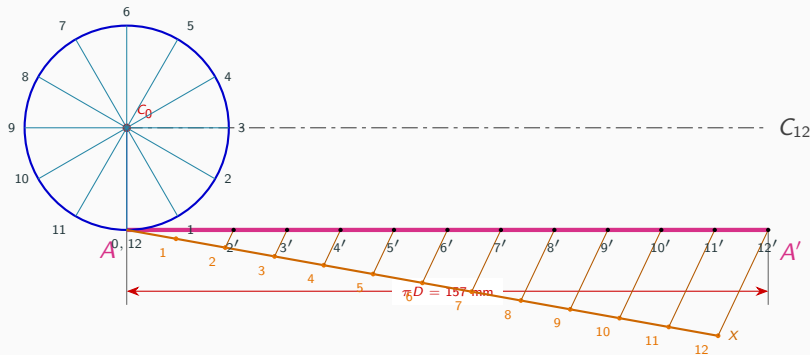
### Step 3: Divide Directing Line - Transfer Point 3'

3' → Transfer Point 3



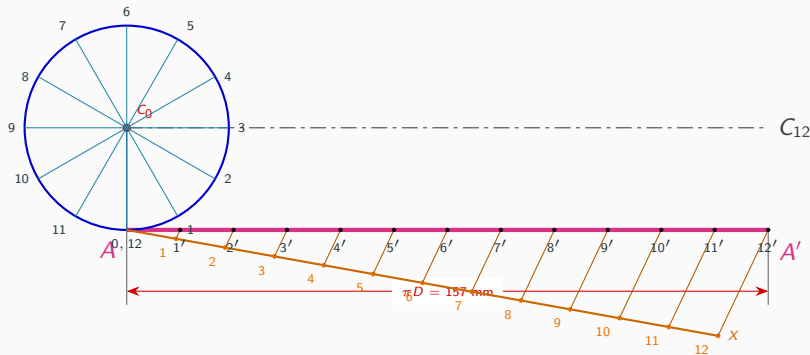
### Step 3: Divide Directing Line - Transfer Point 2'

2' → Transfer Point 2

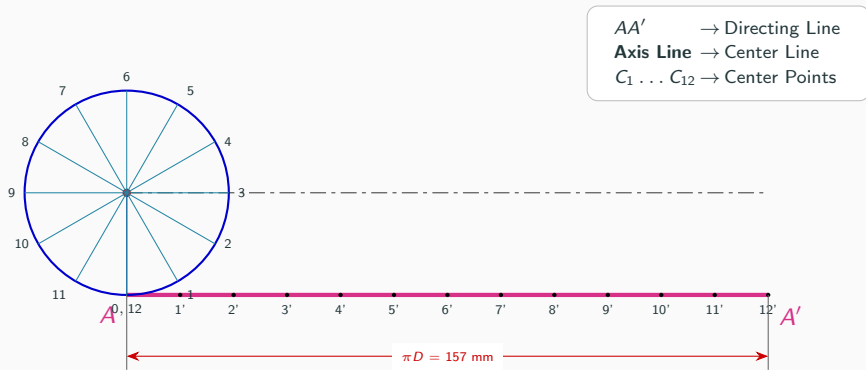


## Step 3: Divide Directing Line - Transfer Point 1'

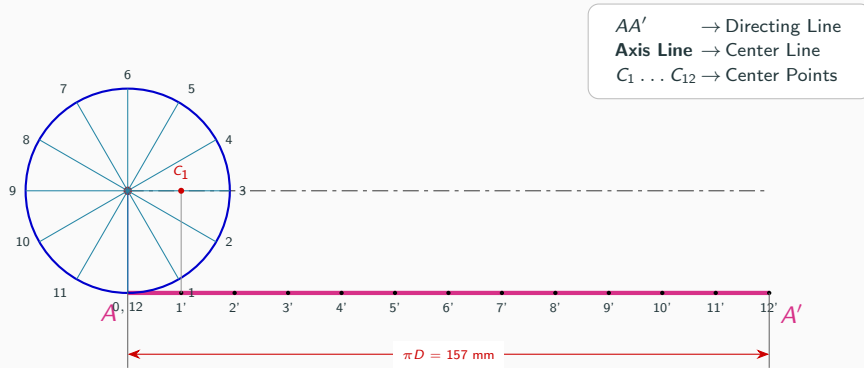
1' → Transfer Point 1



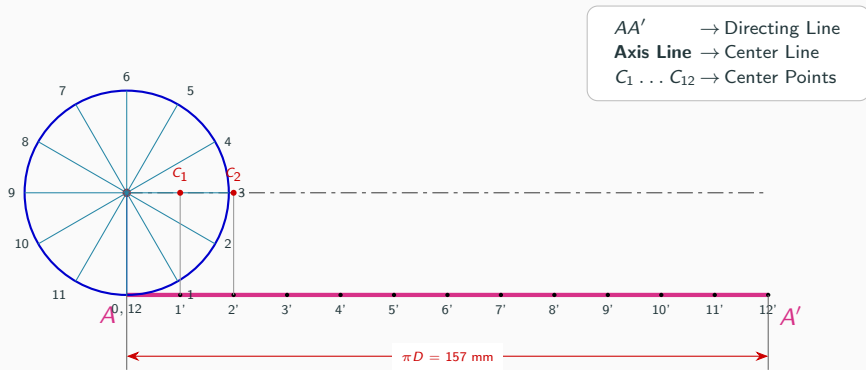
## Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$ (Base)



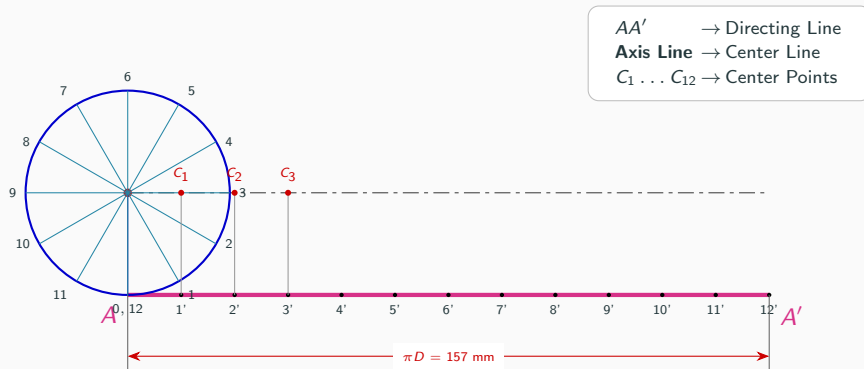
## Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



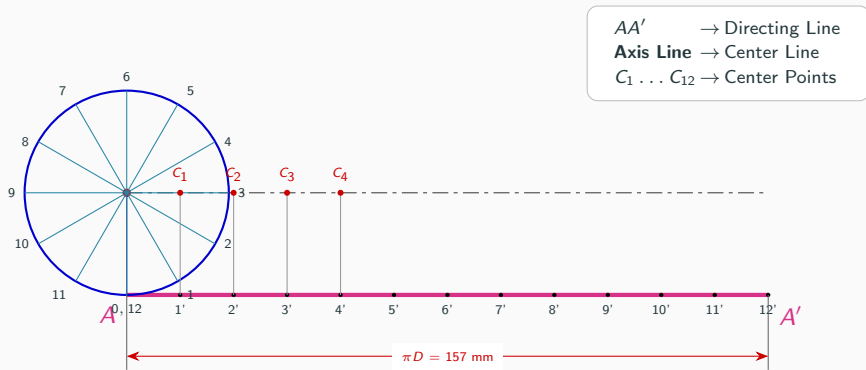
## Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



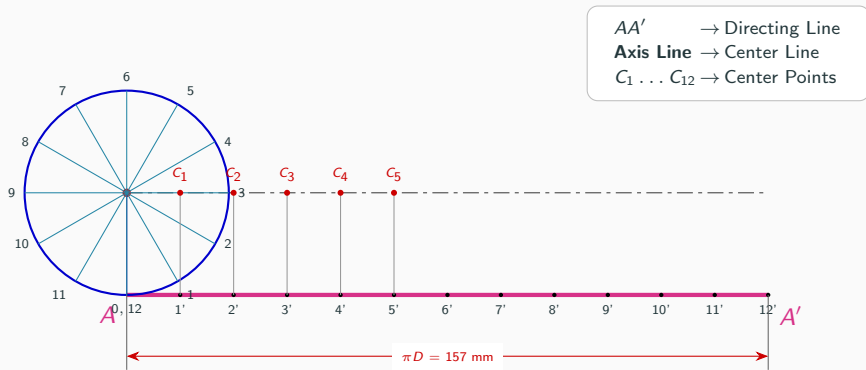
## Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



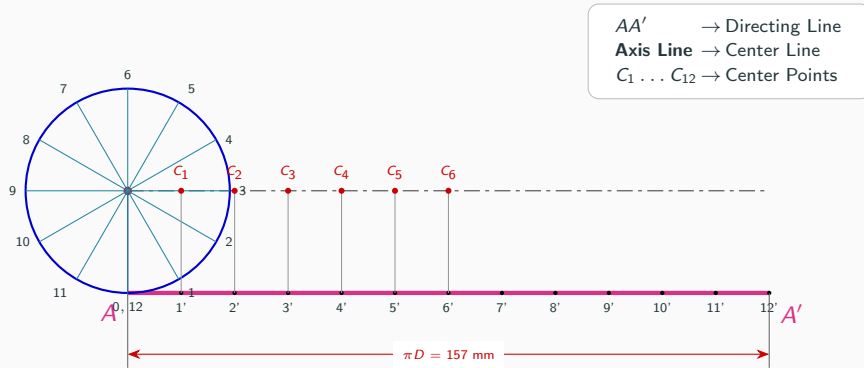
## Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



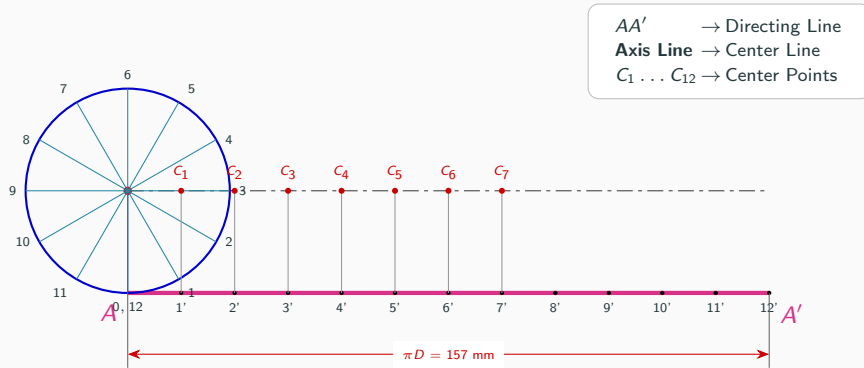
## Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



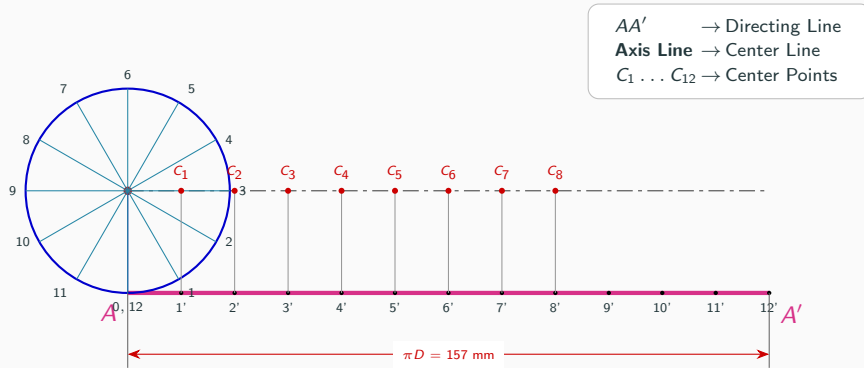
## Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



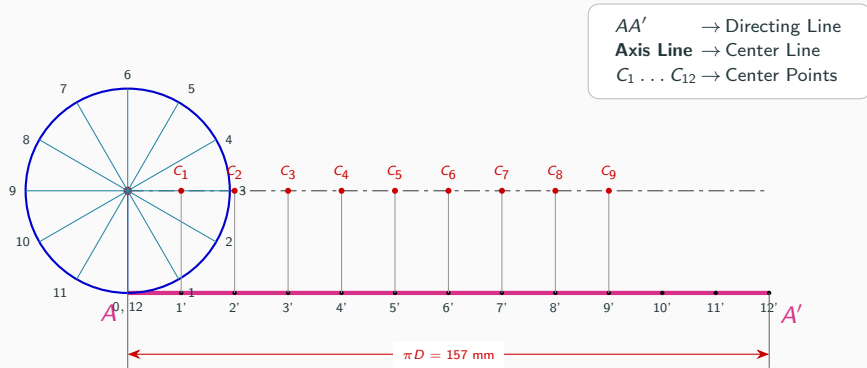
## Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



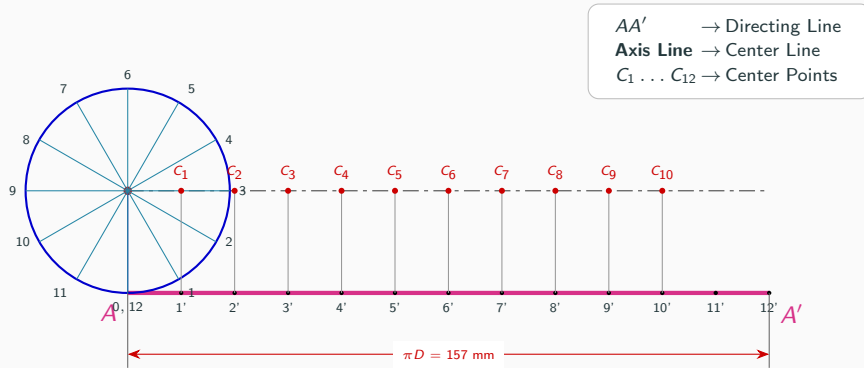
## Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



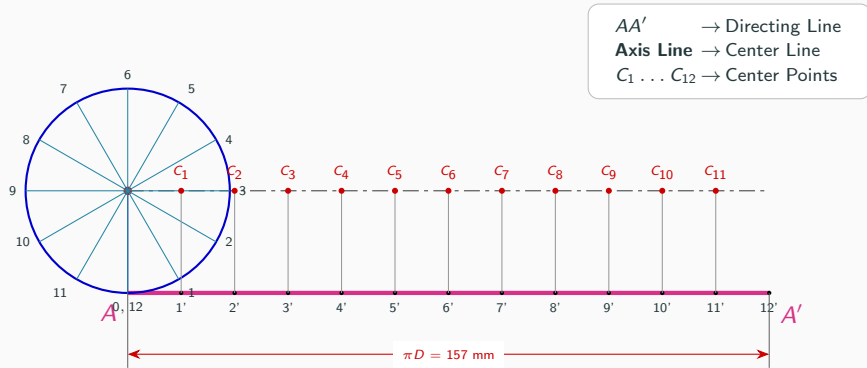
## Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



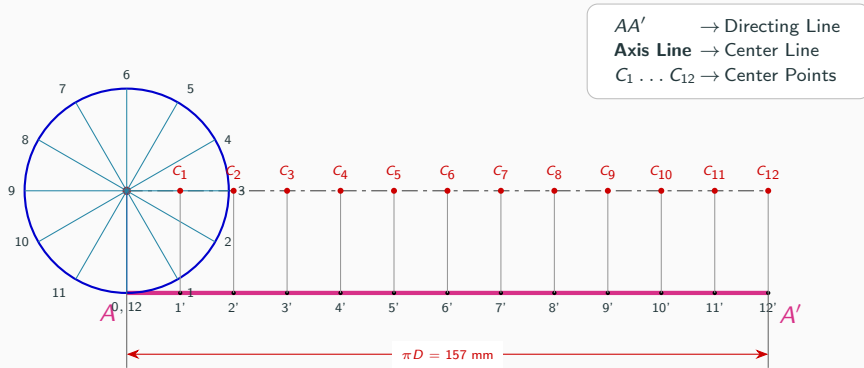
## Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



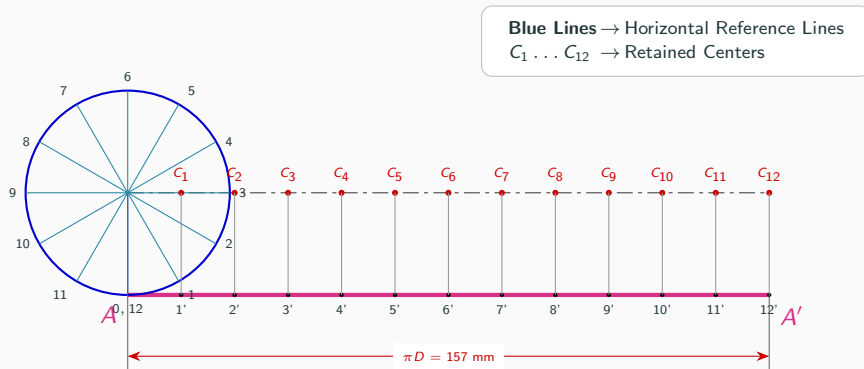
## Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



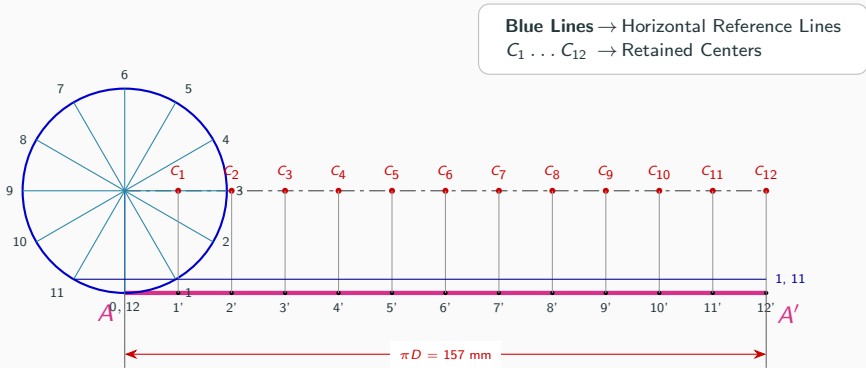
## Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



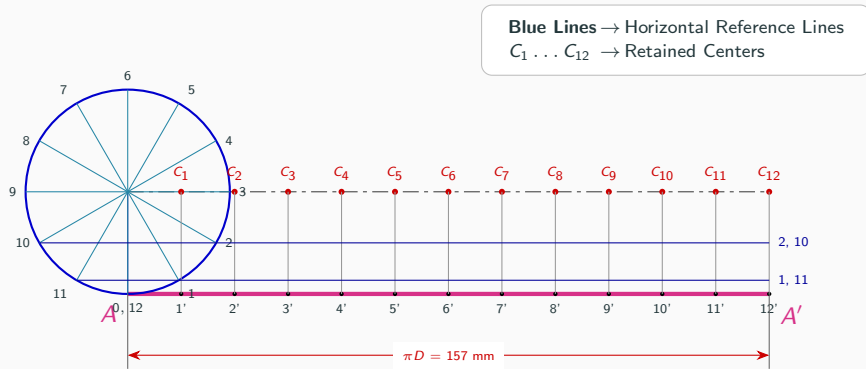
## Step 5: Draw Horizontal Reference Lines (Base)



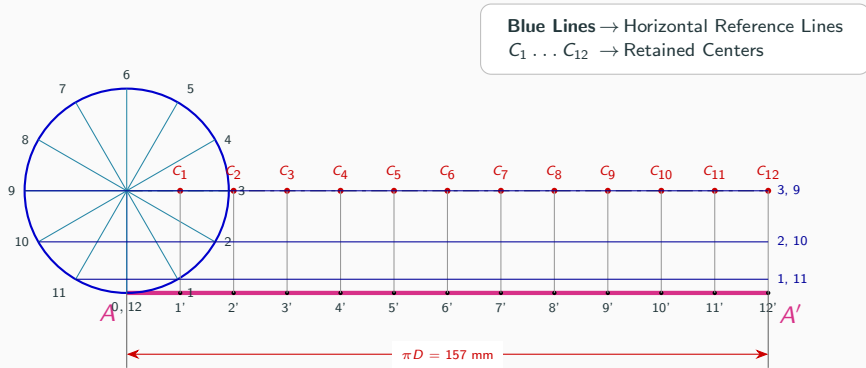
### Step 5: Draw Horizontal Reference Lines



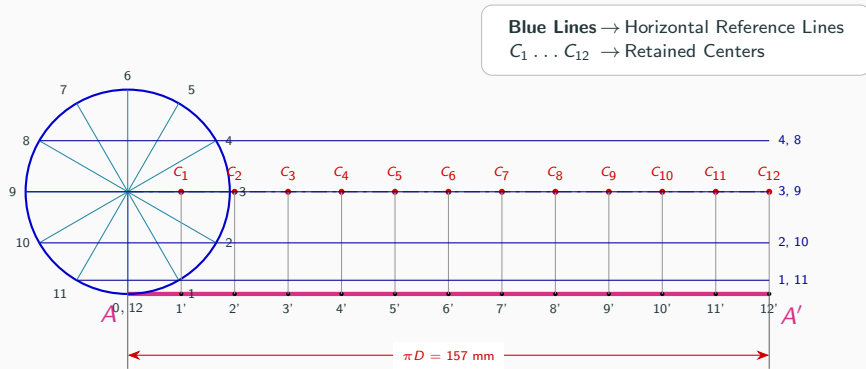
## Step 5: Draw Horizontal Reference Lines



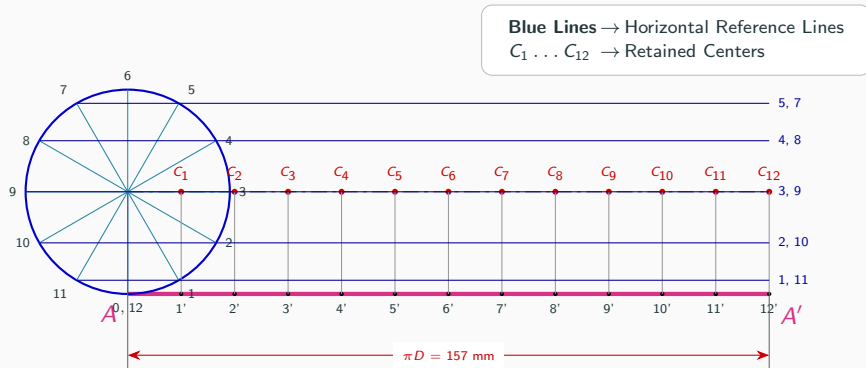
### Step 5: Draw Horizontal Reference Lines



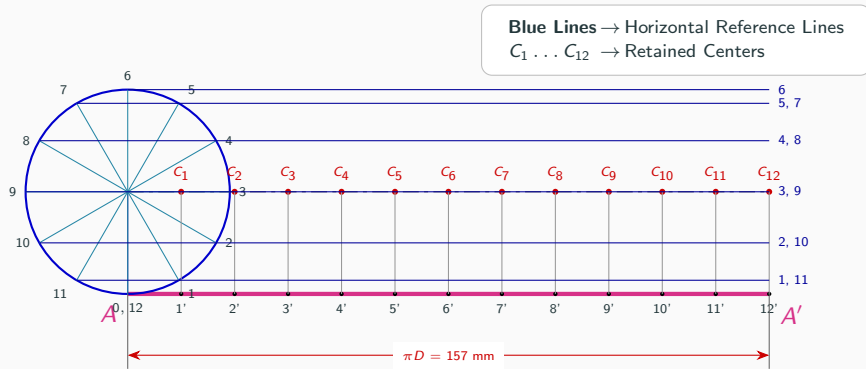
## Step 5: Draw Horizontal Reference Lines



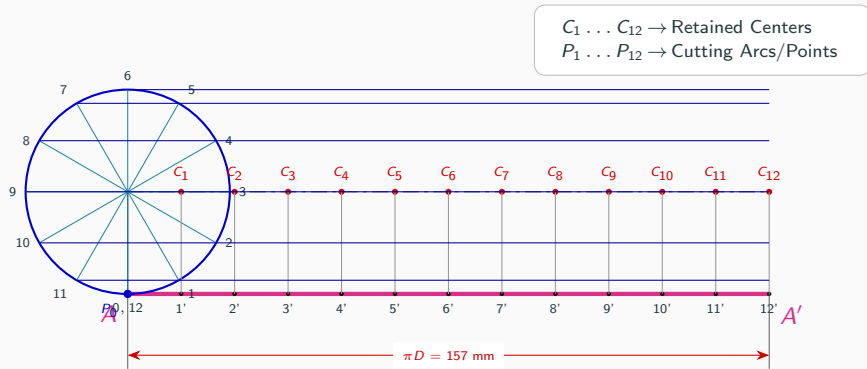
## Step 5: Draw Horizontal Reference Lines



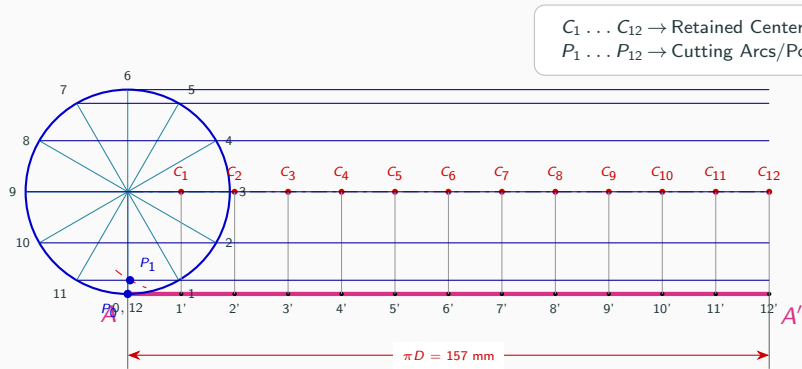
## Step 5: Draw Horizontal Reference Lines



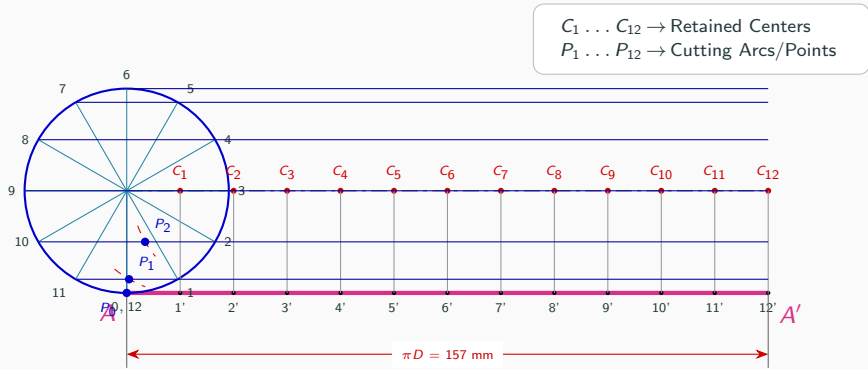
## Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate $P_0$



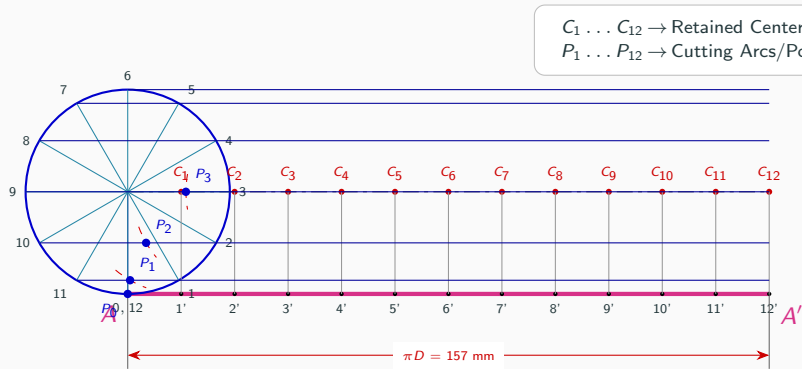
## Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate $P_1$



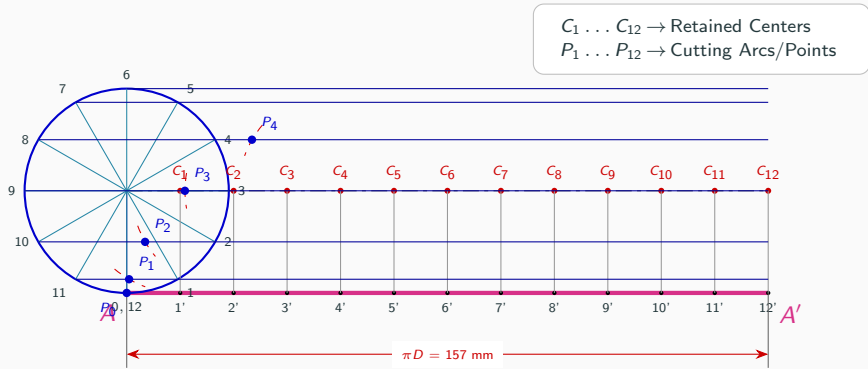
### Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate $P_2$



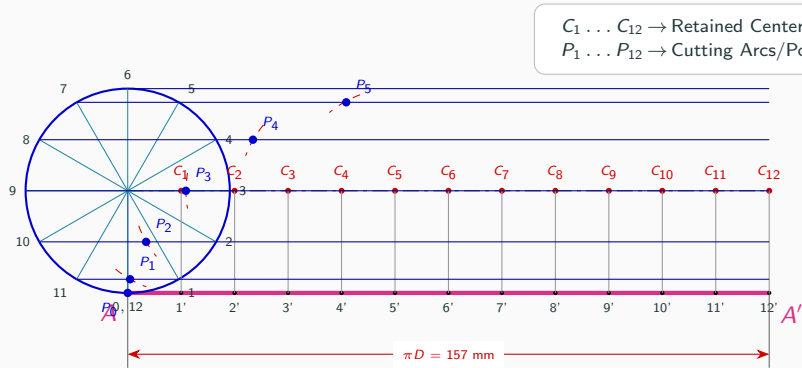
## Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate $P_3$



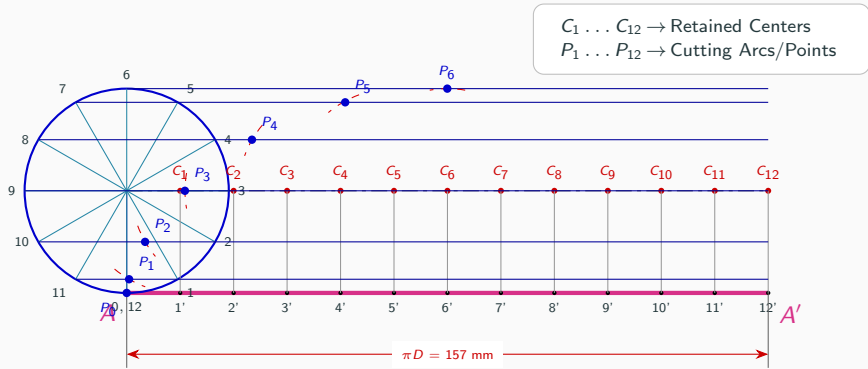
### Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate $P_4$



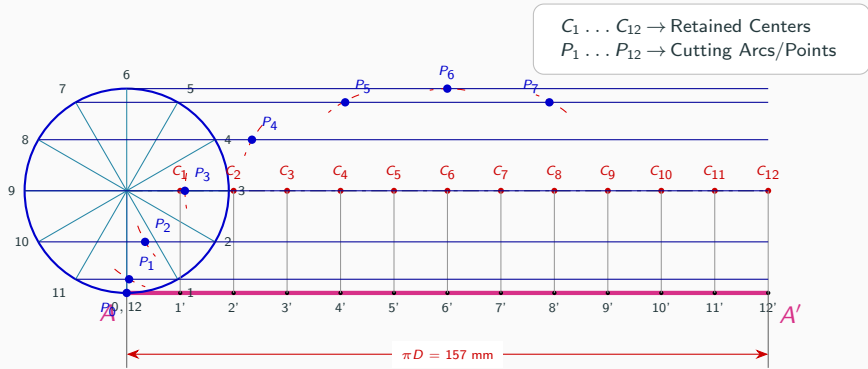
## Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate $P_5$



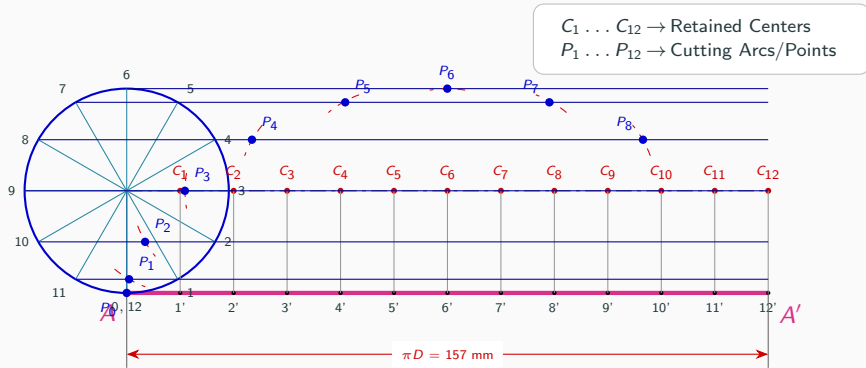
### Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate $P_6$



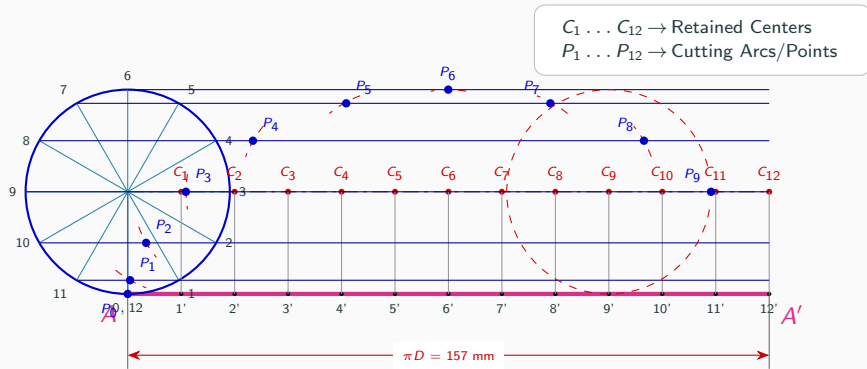
### Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate $P_7$



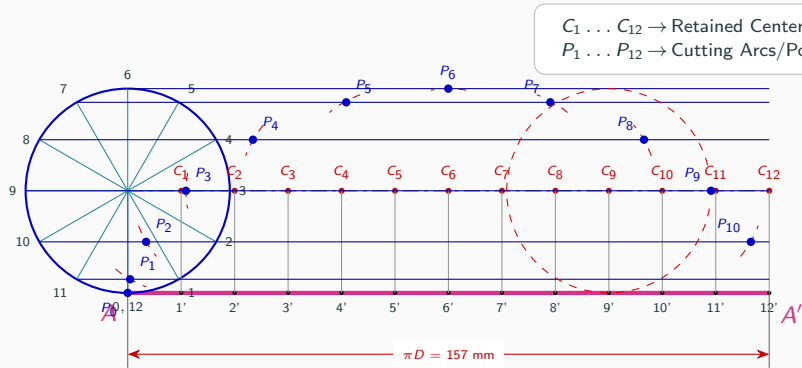
### Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate $P_8$



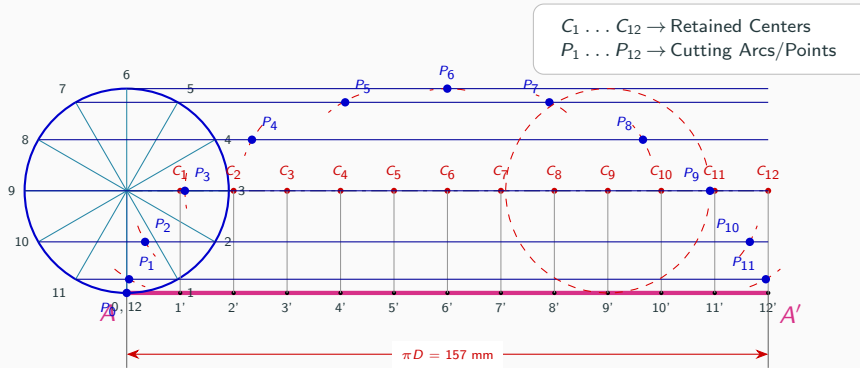
## Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate $P_9$



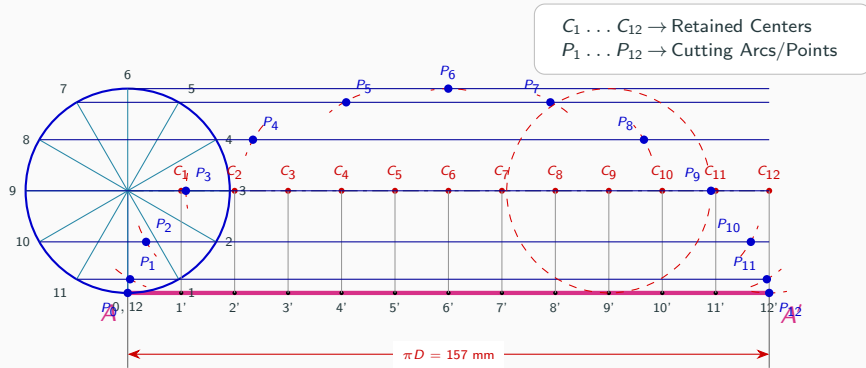
## Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate $P_{10}$



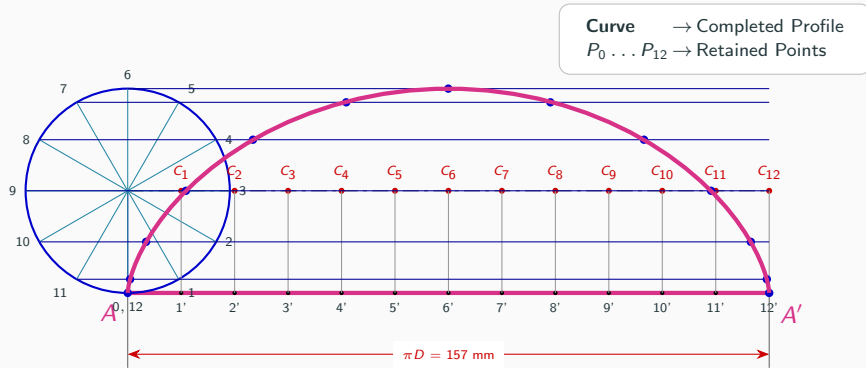
### Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate $P_{11}$



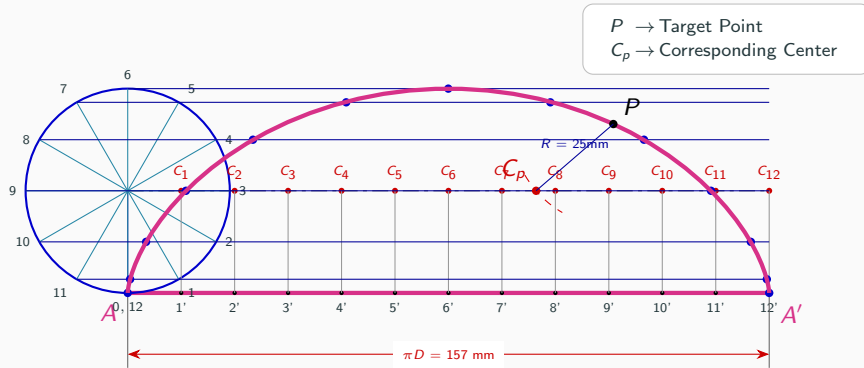
## Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate $P_{12}$



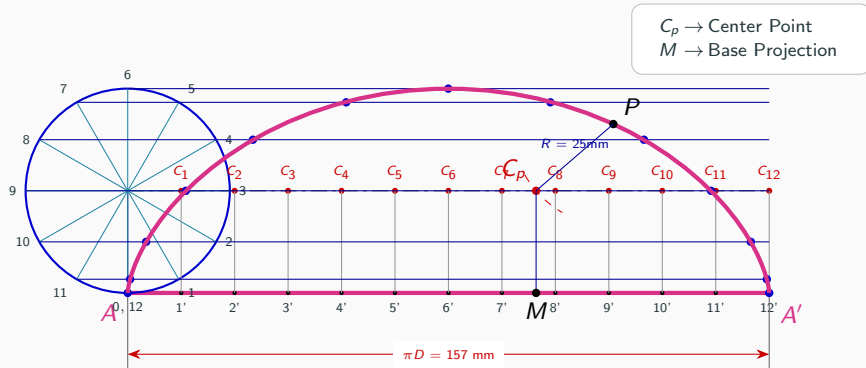
## Step 7: Smoothly Join Points to Form Cycloid Curve



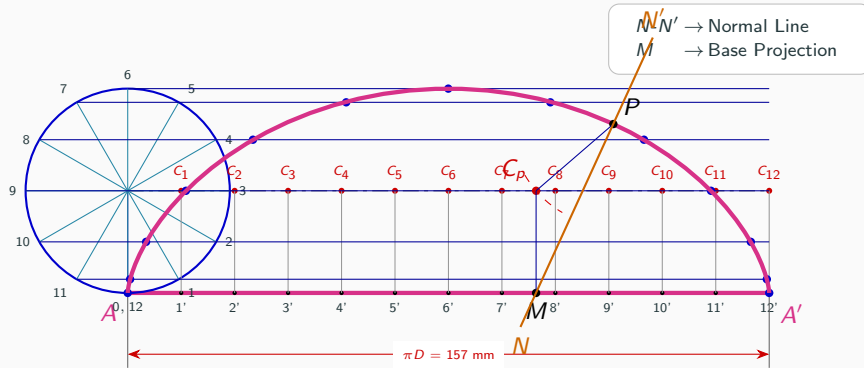
## Step 8: Mark Point $P$ (Right Side) & Locate Center $C_p$



## Step 9a: Project $C_p$ onto Directing Line and Mark $M$



## Step 9b: Construct Normal Line $N-N'$ through $M$ and $P$



## Step 10: Construct Tangent Line $T-T'$ & Final Result

